

AI Usage Note: Quantitative Research Assignment

Candidate Name: [Shourya Singh]

Project: NIFTY 50 Crash Recovery Quantitative Study

AI Collaborator Used: Gemini (Google) as a technical writing and research sparring partner

1. Overview of AI Integration

In the development of this research study, AI tools were utilized strictly as an interactive collaborator and sounding board. The core analytical decisions, mathematical formulas, hypothesis design, and interpretation of quantitative results were directed and validated by the author. AI served to accelerate the coding pipeline, refine prose, and enforce academic and industrial rigor in line with quantitative finance standards.

2. Specific Areas of Collaboration

- **Code Structuring & Debugging:** AI assisted in drafting modular Python scripts for data ingestion via `yfinance`, calculating forward returns while avoiding look-ahead bias (using `shift(-1)` for next-day Open execution), and vectorizing performance metrics.
- **Statistical Implementation:** AI provided syntactic guidance on implementing Welch's T-Tests using `scipy.stats` to evaluate the statistical significance (p-values) of the observed return distributions against market baselines.
- **Research Note Formatting:** AI helped structure the narrative flow of the 2-page Research Note to ensure it met the rigorous communication standards expected in quantitative research (e.g., maintaining an objective tone when presenting a negative result).

3. Human Oversight and Verification

- **Validation of Results:** All outputs, including the discovery of 94 crash events at the -2.0% threshold and the subsequent p-values ($p = 0.5193$), were independently verified through code execution in Google Colab.
- **Critical Thinking & Conclusion:** The primary intellectual conclusion—that the observed recovery pattern is statistically insignificant and likely represents random market noise—was upheld by the author despite initial positive win-rate heuristics. The decision to

reject the hypothesis aligns with rigorous quantitative practices rather than blindly accepting backtest optimization.

4. Reflection

Collaborating with an AI agent significantly compressed the engineering and documentation timeline, allowing focus to shift toward deeper statistical testing (such as threshold sensitivity and out-of-sample chronological splits) rather than boilerplate syntax generation.